House Price Prediction
Regression Analysis Summary — Findings & Recommendations

Objective
This project set out to build a regression model that predicts house prices from property features like area, room counts, location-adjacent attributes, and amenities — and to identify which of those features most strongly drive price, using a real-world housing dataset of 545 properties.

Methodology

- Data preparation: Cleaned the dataset (verified no missing values or duplicates) and one-hot encoded all categorical features (e.g. mainroad, airconditioning, furnishing status).
- Modeling: Split the data 80/20 into training and test sets, then trained two models: Linear Regression and a Random Forest Regressor.
- Evaluation: Compared both models using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

Model Performance

Model	MAE (₹)	RMSE (₹)	R^2 Score
Linear Regression	9,70,043	13,24,507	0.653
Random Forest	10,13,969	13,98,116	0.613

Linear Regression slightly outperformed Random Forest on this dataset, indicating the price relationship is largely linear at this sample size.

Key Insights

- Top price drivers: Area is the single strongest predictor of price, followed by bathrooms, air conditioning, and number of stories. Bedroom count has a surprisingly weak influence by comparison.
- Accuracy: The model explains about 65% of the variation in price ($R^2 \approx 0.65$), with predictions typically within roughly 20% of the actual value — a solid baseline, though not precise enough for final-sale pricing decisions.
- Notable finding: The more complex Random Forest model did not outperform the simpler Linear Regression model, reinforcing that added model complexity isn't automatically better, especially on a modest-sized dataset.

Recommendation
Real estate businesses should prioritize area, bathroom count, and air conditioning when pricing and marketing listings, since these consistently drive the largest share of value — and should treat bedroom count as a secondary factor rather than a primary pricing lever.